

CML — Pathophysiology (Philadelphia / BCR-ABL1)



1. t(9;22) chromosomes

WHAT YOU SEE

Chromosome 9 (blue, ABL1) and chromosome 22 (red, BCR) swapping arms

WHAT IT ENCODES

CML is defined by the reciprocal translocation t(9;22)(q34;q11) that fuses the BCR gene on chromosome 22 with the ABL1 proto-oncogene on chromosome 9. The breakpoint in BCR (major breakpoint cluster region, M-bcr) joins to ABL1 exon 2, creating a chimeric BCR-ABL1 oncogene. This single genetic lesion is necessary and (in chronic phase) sufficient to cause CML.

BOARD TRAP

Know the translocation pairings cold: t(9;22) = CML (and Ph+ ALL); t(15;17) PML-RARA = APL (M3); t(8;14) MYC = Burkitt; t(8;21) and inv(16) = good-prognosis AML; t(11;14) cyclin D1 = mantle cell.

2. Philadelphia chromosome

WHAT YOU SEE

Banner reading PHILADELPHIA over the fusion factory

WHAT IT ENCODES

The shortened derivative chromosome 22 that carries the BCR-ABL1 fusion is the Philadelphia (Ph) chromosome — the first consistent cytogenetic abnormality ever linked to a human cancer and the diagnostic hallmark of CML. It is found in essentially all metaphases by karyotype at diagnosis.

BOARD TRAP

The Ph chromosome is the SMALL derivative chromosome 22, not chromosome 9. Also: Ph positivity is not unique to CML — it appears in ~25% of adult ALL (worse prognosis) and rarely in AML.

3. BCR::ABL1 fusion

WHAT YOU SEE

Red fusion bar labeled BCR::ABL1 with >90%

WHAT IT ENCODES

The BCR-ABL1 fusion is detectable in >90-95% of CML cases and is the driver oncogene. Diagnosis requires molecular confirmation of BCR-ABL1 (by FISH or RT-PCR), not just morphology. The dimerization domain contributed by BCR forces the ABL1 kinase into a constitutively active state.

BOARD TRAP

A normal karyotype (Ph-negative on cytogenetics) does NOT exclude CML — a cryptic/masked BCR-ABL1 fusion can still be present, so always confirm with FISH or PCR before excluding the diagnosis.

4. Constitutive KINASE

WHAT YOU SEE

Glowing KINASE core radiating energy

WHAT IT ENCODES

The fusion protein has a constitutively active ABL1 TYROSINE kinase domain that autophosphorylates and phosphorylates downstream substrates without any upstream signal. This drives unregulated proliferation, resistance to apoptosis, and decreased stromal adhesion. TKIs (imatinib etc.) work by occupying the ATP-binding pocket and locking the kinase off.

BOARD TRAP

BCR-ABL1 is a TYROSINE kinase, not a serine/threonine kinase — this is why tyrosine kinase inhibitors (not mTOR/serine-threonine inhibitors) are the treatment.

5. Downstream pathways

WHAT YOU SEE

Pipes labeled RAS, MAPK, STAT, PI3K, MYC leaving the kinase

WHAT IT ENCODES

Active BCR-ABL1 fires multiple oncogenic cascades simultaneously: RAS/MAPK (proliferation), JAK2-STAT5 (survival/anti-apoptosis via BCL-XL), and PI3K/AKT (growth and survival), plus MYC upregulation. The net effect is expansion of the myeloid clone with genomic instability that predisposes to additional mutations.

6. Myristoyl pocket lost

WHAT YOU SEE

Broken lock labeled MYRISTOYL LOST

WHAT IT ENCODES

Normal ABL1 is autoinhibited by an N-terminal myristoyl group that folds into a regulatory pocket. The BCR fusion removes this cap, so the kinase loses its 'off switch' and stays permanently active. Asciminib is a STAMP inhibitor that restores inhibition by binding this allosteric myristoyl pocket — a mechanism distinct from ATP-competitive TKIs.

BOARD TRAP

Asciminib targets the myristoyl pocket allosterically (STAMP), so it can work where ATP-pocket mutations like T315I cause resistance to conventional TKIs.

7. p210 (e13a2/e14a2)

WHAT YOU SEE

Orange p210 barrel labeled e13a2/e14a2 with PCR probe

WHAT IT ENCODES

p210 (transcripts e13a2/b2a2 and e14a2/b3a2, from the major breakpoint M-bcr) is the classic CML fusion protein, present in ~95% of cases. These are the standard transcripts quantified by RT-PCR on the International Scale for monitoring response and are required to attempt treatment-free remission.

BOARD TRAP

Treatment-free remission attempts require a 'typical' p210 transcript (e13a2/e14a2) that is quantifiable on the International Scale — atypical transcripts can't be monitored reliably.

8. p190 = ALL

WHAT YOU SEE

Smaller p190 barrel labeled ALL, worse in CML

WHAT IT ENCODES

p190 (e1a2, from the minor breakpoint m-bcr) is the smaller fusion typical of Ph-positive ALL. When p190 occurs in CML it is uncommon and associated with monocytosis and a worse, more aggressive course.

BOARD TRAP

p190 in a CML patient is a RED FLAG (more aggressive disease / monocytosis), not reassuring. p190 = think Ph+ ALL; p210 = think CML.

9. e13a3/e14a3 false neg

WHAT YOU SEE

Indolent barrel labeled e13a3/e14a3 with FALSELY NEGATIVE

WHAT IT ENCODES

Atypical/rare fusion transcripts (e.g. e13a3, e14a3, e1a3, e19a2/p230) may not be picked up by standard PCR primers targeting p210/p190, producing a falsely negative or unquantifiable result. If clinical suspicion is high but PCR is negative, request atypical-transcript testing or FISH.

BOARD TRAP

A negative quantitative PCR does not exclude CML if an atypical transcript is present — the standard primer set can miss it.

10. Lineages involved

WHAT YOU SEE

List: myeloid, erythroid, megakaryo, monocyte; B-cell/T-cell crossed out

WHAT IT ENCODES

CML arises from a transformed pluripotent hematopoietic STEM cell, so the Ph clone is present across granulocytic, erythroid, megakaryocytic, and monocytic lineages (and sometimes B-lymphoid). This multilineage involvement explains why the peripheral smear shows the full myeloid maturation spectrum plus thrombocytosis.

11. Second hit / progression

WHAT YOU SEE

SECOND HIT lightning bolt over normal crowd

WHAT IT ENCODES

BCR-ABL1 alone causes chronic phase; progression to accelerated/blast phase requires additional 'second hits' (genomic instability driving mutations in TP53, RUNX1, ASXL1, loss of p16/CDKN2A, etc.). This stepwise acquisition is why untreated CML transforms over a few years and why TKIs — by suppressing the clone — prevent accumulation of these hits.

12. Clonal evolution markers

WHAT YOU SEE

Bottom track: +8, iso17, 20q-, 3q26.2, double Ph → TRANSFORMATION

WHAT IT ENCODES

Additional cytogenetic abnormalities (ACAs) in the Ph clone signal clonal evolution and impending transformation. High-risk ACAs include +8 (trisomy 8), i(17q)/iso17, a second Ph (double Ph), and 3q26.2 (EVI1/MECOM) rearrangements. Their appearance on serial karyotype is itself an accelerated-phase criterion.

BOARD TRAP

Major-route ACAs (+8, i(17q), double Ph, 3q26.2) at diagnosis predict poorer response and are used to define accelerated phase — they prompt consideration of a 2nd-gen TKI and closer monitoring.

13. Incidence 2/100,000

WHAT YOU SEE

2 IN 100,000 label under the crowd

WHAT IT ENCODES

CML is a chronic myeloproliferative neoplasm with an incidence of ~1-2 per 100,000 per year, making up ~15% of adult leukemias. Median age at diagnosis is ~55-65 years with a slight male predominance (M:F ~1.6:1). Ionizing radiation is the only well-established risk factor.